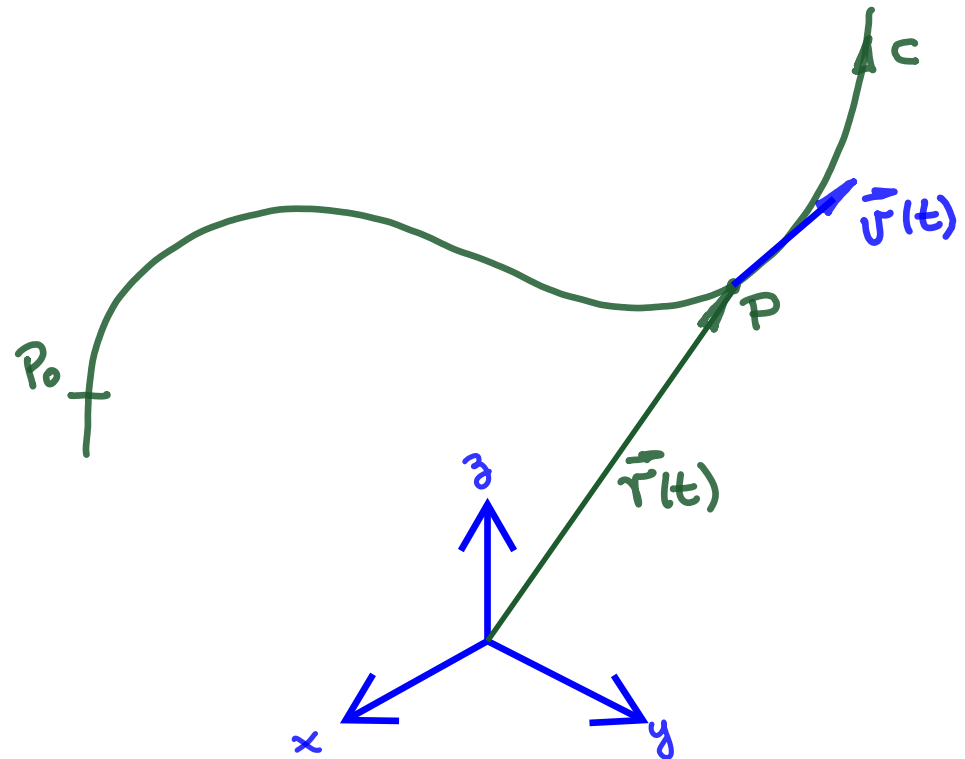


Tangential and Normal Components of Acceleration

Trim Ch 11, S. 11.13

Outline

1. Tangential and Normal Components of Acceleration
2. A different formula for Curvature



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Tangential and Normal Components of Acceleration

For a body traveling in a curve defined by the path $\vec{r}(t)$ that is accelerating, the acceleration is given by the derivative of the velocity vector.

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt} \left(\frac{d\vec{r}}{dt} \right)$$

Using the chain rule, we can write the velocity vector as

Tangential and Normal Components of Acceleration

Acceleration components in terms of “t”

Tangential scalar component a_T :

Normal scalar component a_N :

Notice that a_N makes the particle/body follow a curved path with direction parallel to the instantaneous radius of curvature “ ρ ” (centripetal force).



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Another formula for the Normal Component of Acceleration

Another formula for the normal component of the acceleration, which does not require us to calculate the curvature ($k = \frac{1}{|\vec{v}|} \frac{d\hat{T}}{dt}$), can be obtained from using the fact that,

$$|\vec{a}|^2 = \vec{a} \cdot \vec{a}$$

Another Formula for Curvature

Find a different formula for curvature using the following formulas for velocity,

$$\vec{v} = \frac{ds}{dt} \hat{T}$$

and for acceleration,

$$\vec{a} = \frac{d^2s}{dt^2} \hat{T} + k \left(\frac{ds}{dt} \right)^2 \hat{N}$$

Solution:

Another Formula for Curvature

Another Formula for Curvature

Problem

A particle has a trajectory defined by,

$$x = t, \quad y = t^2, \quad z = t^2$$

where “ t ” is time, $t \geq 0$

Find the **normal and tangential components** of its velocity and acceleration and express them in terms of $\hat{T}$ and $\hat{N}$, **without calculating neither $\hat{T}$ nor $\hat{N}$**

Solution

Problem

Problem

Summary

- The tangential component of acceleration gives us information on the rate of change of speed
- The normal component of acceleration is related to the centripetal force
- The components of the acceleration lie in the plane determined by $\hat{T}$ and $\hat{N}$, or the osculating plane